Computer Networks Lab Report

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EXPERIMENT 1: IMPLEMENTATION OF BASIC NETWORK TOPOLOGY USING CISCO PACKET TRACER

1. AIM

To build a simple network by linking two PCs to a switch in Cisco Packet Tracer and to confirm that they can communicate using the ping command.

2. SOFTWARE AND DEVICES REQUIRED

Category	Item	Quantity
Software	Cisco Packet Tracer	1
End device	PC-PT (generic PC)	2
Network device	Cisco 2950-24 switch	1
Cable	Copper straight-through	2

3. THEORY

A simple network links end devices through a switch so that they can exchange data. The switch records the MAC address of every attached device and sends each frame only out of the port that leads to the destination. Devices on the same network must have IP addresses that share the same network portion, and reachability is checked with the ping command, which transmits ICMP echo requests and waits for the echo replies. Simulation mode in Packet Tracer displays the journey of every packet one step at a time.

4. PROCEDURE

1. Start Cisco Packet Tracer and drag two PCs (PC0 and PC1) and one 2950-24 switch (Switch0) into the logical workspace.
2. Link PC0 and PC1 to Switch0 with copper straight-through cables.
3. On each PC, go to Desktop and then IP Configuration, choose Static and type in the IPv4 address and subnet mask.
4. Wait for the link indicators to change to green.
5. Change to simulation mode, send a simple PDU from PC0 to PC1 and follow its path.
6. Open the Command Prompt on PC0 and execute ping 192.168.10.2.

5. IP ADDRESS CONFIGURATION

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.10.1	255.255.255.0
PC1	FastEthernet0	192.168.10.2	255.255.255.0

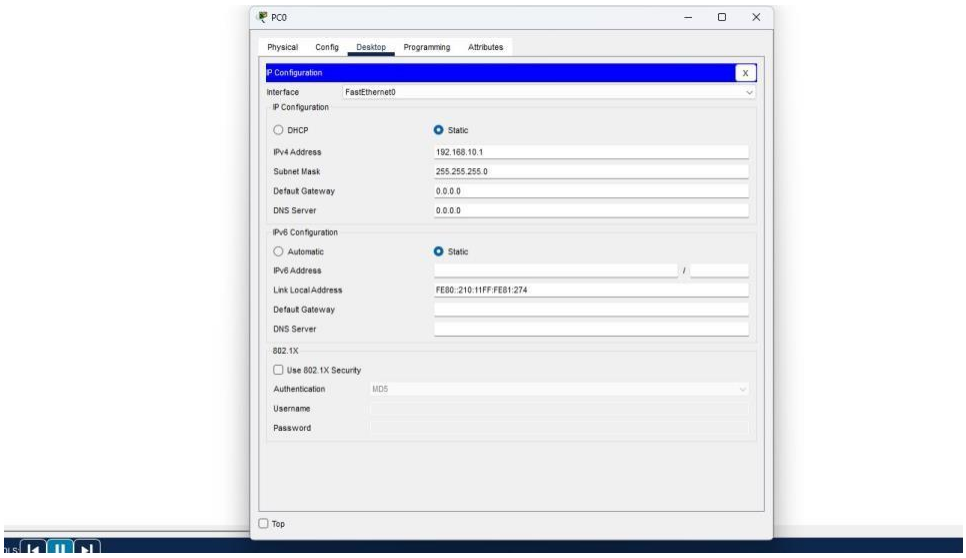


Figure 1. IP configuration of PC0

6. OBSERVATION

The PDU originating at PC0 arrived at Switch0 and was then passed on to PC1. The ping from PC0 to 192.168.10.2 transmitted 4 packets and got 4 replies back with 0% loss; each reply came in under 1 ms with TTL=128.

7. OUTPUT

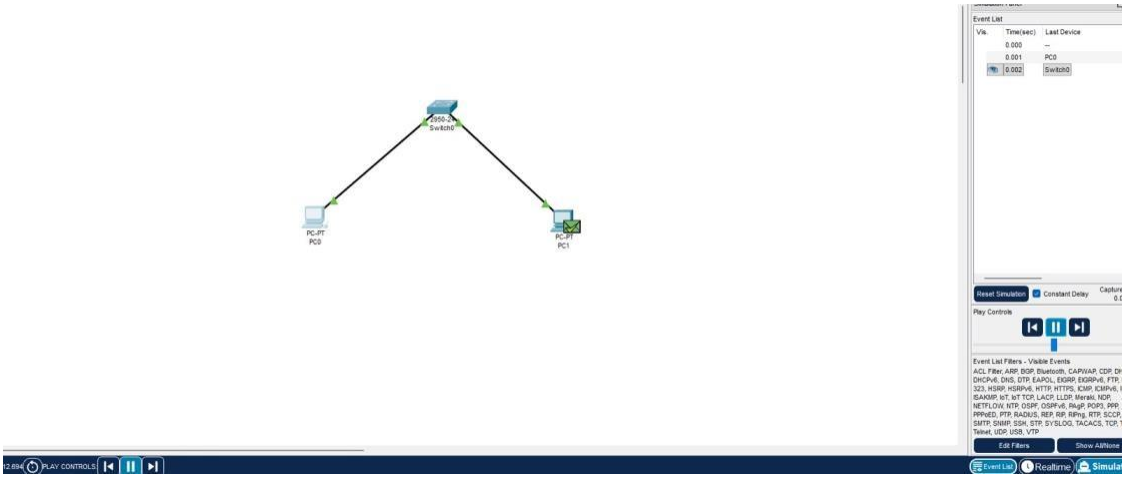


Figure 2. Basic network in simulation mode with a PDU from PC0 to PC1 through Switch0

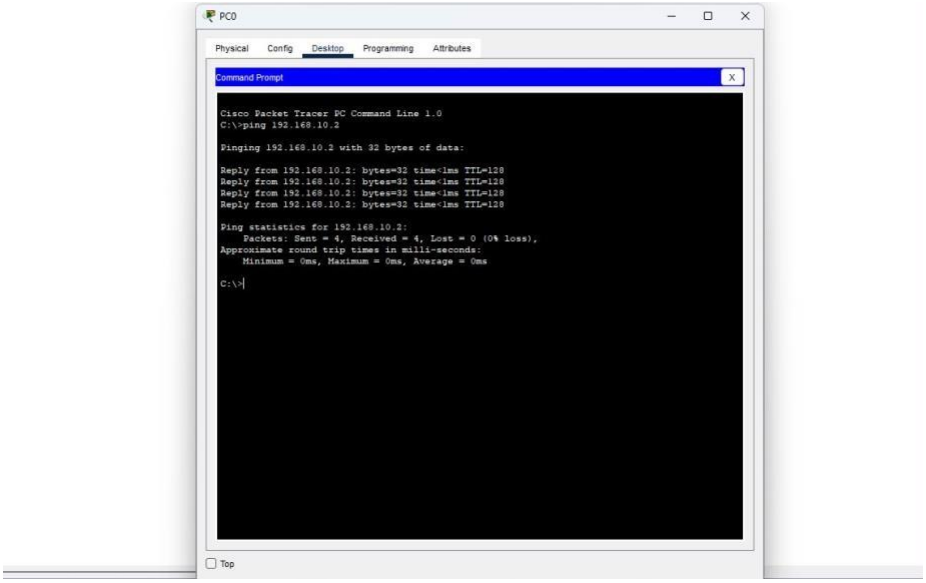


Figure 3. Command prompt of PC0 showing ping replies from 192.168.10.2

8. RESULT

The simple network of two PCs joined through a switch was built successfully, and communication between PC0 and PC1 was confirmed using ping.

EXPERIMENT 2: IMPLEMENTATION AND STUDY OF DIFFERENT NETWORK TOPOLOGIES USING CISCO PACKET TRACER

1. AIM

To design, build and examine the point-to-point, bus, star, ring, mesh, tree and hybrid network topologies in Cisco Packet Tracer, and to check connectivity in every topology with the ping command and simulation mode.

2. SOFTWARE AND DEVICES REQUIRED

Category	Item	Quantity
Software	Cisco Packet Tracer	1
End device	PC-PT (generic PC)	As per topology
Network device	Cisco 2950-24 switch	As per topology
Network device	Hub-PT (hub)	1 (hybrid only)
Cable	Copper straight-through	As per topology
Cable	Copper cross-over	As per topology

3. THEORY

A network topology is the layout of the nodes and links that make up a network. The physical topology shows how the devices are cabled, whereas the logical topology shows how data flows between them. The topology chosen determines the cost, performance, fault tolerance and scalability of the network.

Point-to-point, bus, star, ring, mesh, tree and hybrid are the seven topologies covered in this experiment. In Cisco Packet Tracer, straight-through cables connect different types of devices (PC to switch) while cross-over cables connect similar devices (switch to switch, PC to PC). Reachability is tested by sending ICMP echo requests with the ping command, and the route taken by a packet is followed in simulation mode.

4. IMPLEMENTATION OF NETWORK TOPOLOGIES

4.1 Point-to-Point Topology

Aim: To build a point-to-point topology between two PCs and confirm connectivity using ping.

Theory: A point-to-point topology connects exactly two devices over a dedicated link, so the entire bandwidth of that link belongs to the two devices. It is the easiest topology to set up and secure, but it cannot grow beyond two nodes unless additional devices are introduced.

Devices Required

Device / Item	Type / Model	Quantity
PC-PT	End device	2
Copper cross-over cable	Cable	1

Procedure

1. Start Cisco Packet Tracer and place two PCs (PC0 and PC1) in the logical workspace.
2. Choose the copper cross-over cable and join FastEthernet0 of PC0 to FastEthernet0 of PC1.
3. On each PC, open Desktop and then IP Configuration, and set the IP address and subnet mask.
4. Wait for both link indicators to turn green.
5. Open the Command Prompt on PC0 and ping the IP address of PC1.

IP Address Configuration

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.10.1	255.255.255.0
PC1	FastEthernet0	192.168.10.2	255.255.255.0

Screenshot

[Insert point-to-point topology and command prompt screenshot here]

Observation: The two PCs are connected directly without any device in between. Both belong to the 192.168.10.0/24 network, so they are able to exchange packets across the dedicated link.

Result: The point-to-point topology was built and the connection between PC0 and PC1 was confirmed.

4.2 Bus Topology

Aim: To build a bus topology using a linear backbone of switches and confirm connectivity between the end devices.

Theory: In a bus topology every node is attached to one shared backbone. Data transmitted by a node moves along the backbone and is visible to all nodes, but only the intended recipient accepts it. It requires little cable and is inexpensive, yet a break in the backbone divides the network and performance drops as more nodes are attached. In Packet Tracer the backbone is represented by a straight chain of switches.

Devices Required

Device / Item	Type / Model	Quantity
PC-PT	End device	4
Cisco 2950-24 switch	Network device	4
Copper straight-through cable	Cable (PC to switch)	4
Copper cross-over cable	Cable (switch to switch)	3

Procedure

1. Put four PCs (PC0 to PC3) and four 2950-24 switches (Switch0 to Switch3) in the workspace.
2. Join Switch0, Switch1, Switch2 and Switch3 in a straight line with copper cross-over cables to create the backbone.
3. Attach PC0, PC1, PC2 and PC3 to Switch0, Switch1, Switch2 and Switch3 respectively with copper straight-through cables.
4. Give each PC an IP address from the 192.168.10.0/24 network.
5. Change to simulation mode, send a simple PDU from PC0 to PC3 and follow its path.
6. Open the Command Prompt on PC3 and run the ping command.

Screenshot

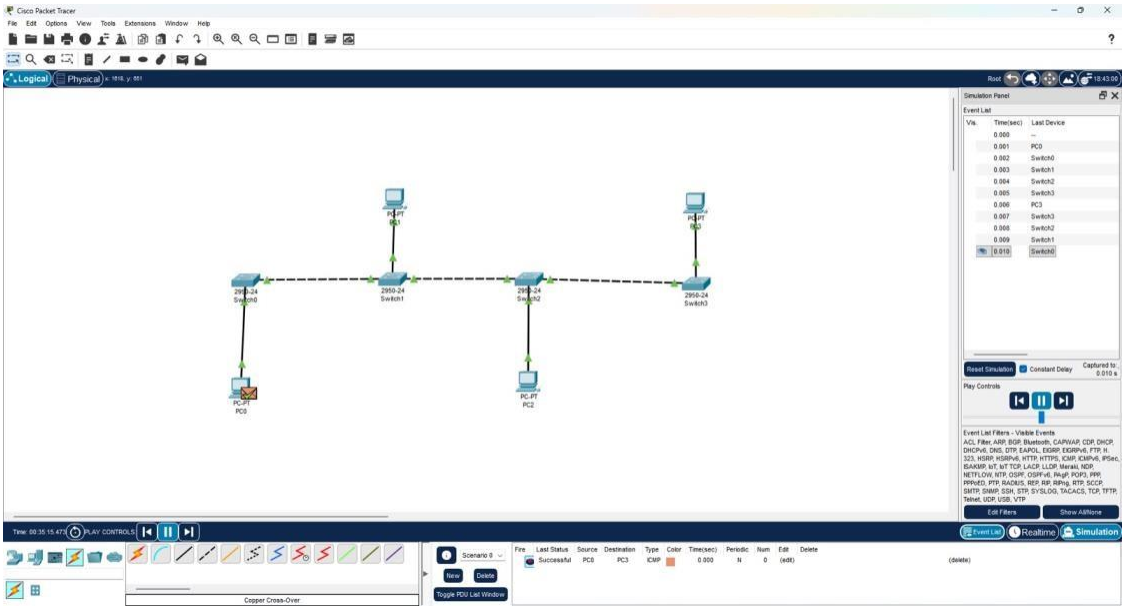


Figure 4. Bus topology in simulation mode with a successful ICMP PDU from PC0 to PC3

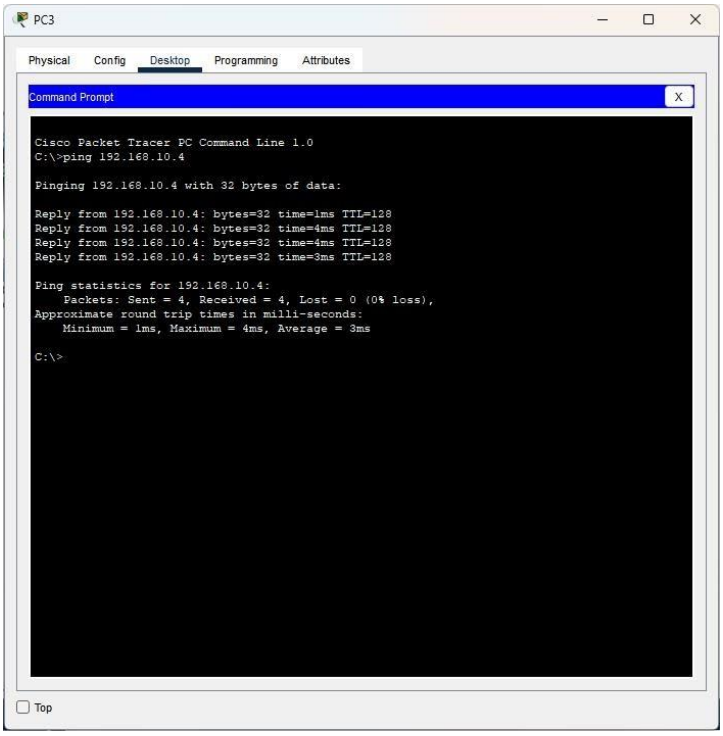


Figure 5. Command prompt of PC3 showing ping replies from 192.168.10.4

Observation: The simple PDU from PC0 to PC3 moved through Switch0, Switch1, Switch2 and Switch3 one after another along the backbone and came back by the same route. The ping test transmitted 4 packets and received 4 replies with 0% loss; the round-trip time varied from 1 ms to 4 ms with an average of 3 ms and TTL=128.

Result: The bus topology was built and end-to-end connectivity across the backbone was confirmed.

4.3 Star Topology

Aim: To build a star topology around a central switch and confirm connectivity between the PCs.

Theory: In a star topology each node is connected to a central device, which here is a switch. All traffic goes through the switch, which sends frames only to the port of the destination. A damaged cable disrupts just one node and new nodes can be added easily, but the central switch is a single point of failure.

Devices Required

Device / Item	Type / Model	Quantity
PC-PT	End device	4
Cisco 2950-24 switch	Network device	1
Copper straight-through cable	Cable	4

Procedure

1. Put one 2950-24 switch (Switch1) at the centre of the workspace and arrange four PCs (PC4 to PC7) around it.
2. Link every PC to the switch with a copper straight-through cable.
3. Set the IP address and subnet mask on each PC.
4. Change to simulation mode, send a simple PDU from PC4 to PC7 and follow its path.
5. Open the Command Prompt on PC4 and ping 192.168.10.4.

IP Address Configuration

Device	Interface	IP Address	Subnet Mask
PC4	FastEthernet0	192.168.10.1	255.255.255.0
PC5	FastEthernet0	192.168.10.2	255.255.255.0
PC6	FastEthernet0	192.168.10.3	255.255.255.0
PC7	FastEthernet0	192.168.10.4	255.255.255.0

Screenshot

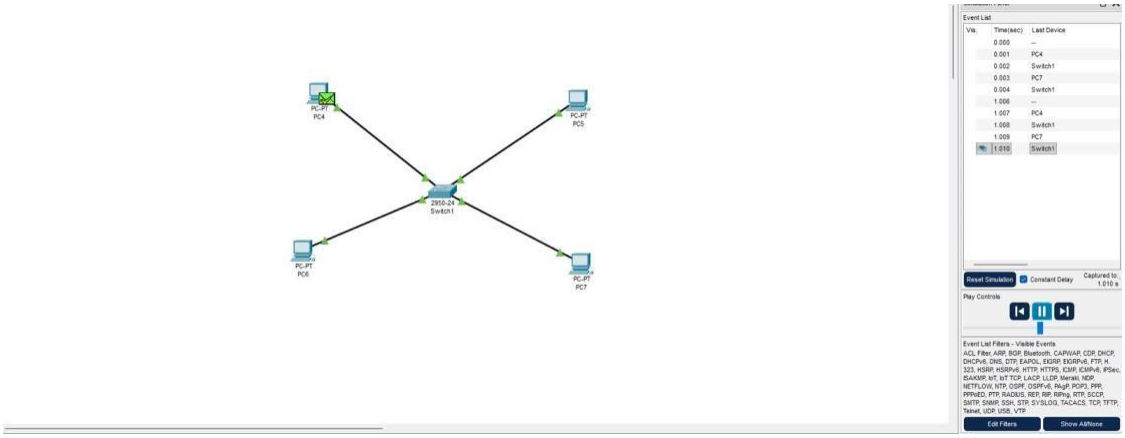


Figure 6. Star topology in simulation mode with a PDU from PC4 to PC7 through Switch1

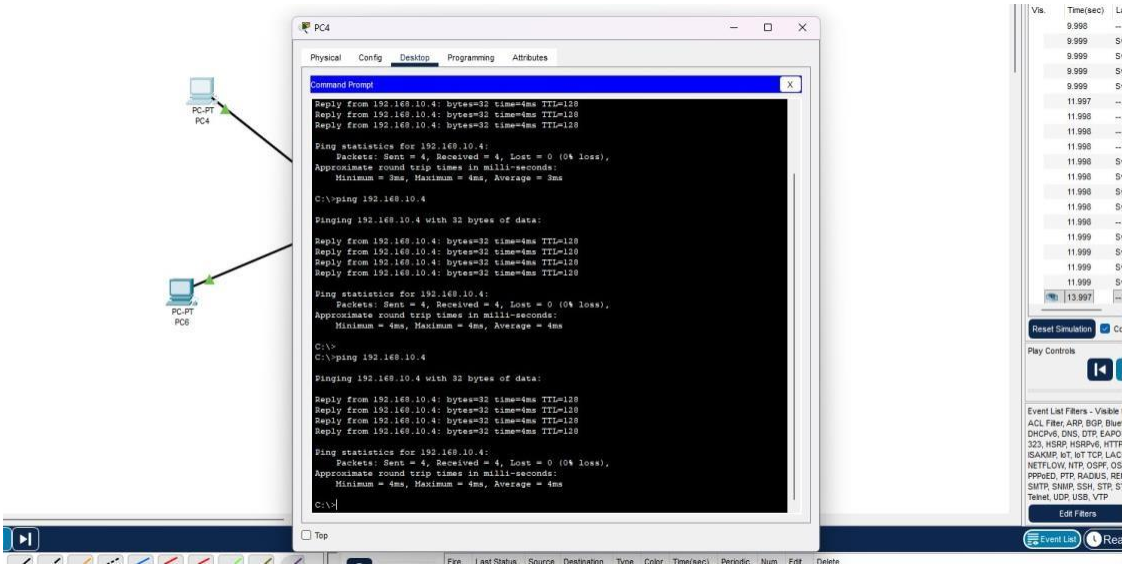


Figure 7. Command prompt of PC4 showing ping replies from 192.168.10.4

Observation: The PDU from PC4 arrived at PC7 by passing only through the central switch, and the reply took the same route back. Every ping test transmitted 4 packets and received 4 replies with 0% loss; the round-trip time was 3 ms to 4 ms and TTL=128.

Result: The star topology was built and connectivity among all the PCs through the central switch was confirmed.

4.4 Ring Topology

Aim: To build a ring topology with four switches and confirm connectivity between the PCs.

Theory: In a ring topology each node is linked to exactly two neighbours, so the nodes together form a closed loop. Data can move in either direction and an alternative path is available if a link breaks. A switched ring produces a Layer 2 loop, so the Spanning Tree Protocol (STP) blocks one link (marked in orange in Packet Tracer) and activates it again when another link goes down.

Devices Required

Device / Item	Type / Model	Quantity
PC-PT	End device	4
Cisco 2950-24 switch	Network device	4
Copper straight-through cable	Cable (PC to switch)	4
Copper cross-over cable	Cable (switch to switch)	4

Procedure

1. Put four 2950-24 switches (Switch0 to Switch3) and four PCs (PC0 to PC3) in the workspace.
2. Join the switches in a closed loop with copper cross-over cables: Switch0 to Switch1, Switch1 to Switch3, Switch3 to Switch2 and Switch2 to Switch0.
3. Attach one PC to every switch with a copper straight-through cable.
4. Set the IP address and subnet mask on each PC.
5. Change to simulation mode, send a simple PDU from PC0 to PC1 and follow its path.
6. Open the Command Prompt on PC3 and ping 192.168.10.4.

IP Address Configuration

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.10.1	255.255.255.0
PC1	FastEthernet0	192.168.10.2	255.255.255.0
PC2	FastEthernet0	192.168.10.3	255.255.255.0
PC3	FastEthernet0	192.168.10.4	255.255.255.0

Screenshot

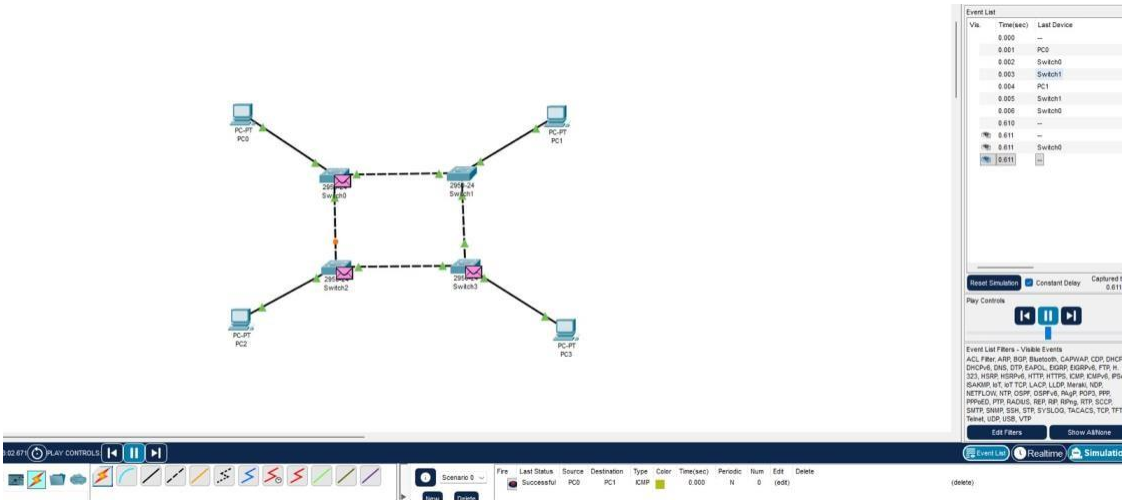


Figure 8. Ring topology in simulation mode with a PDU from PC0 to PC1

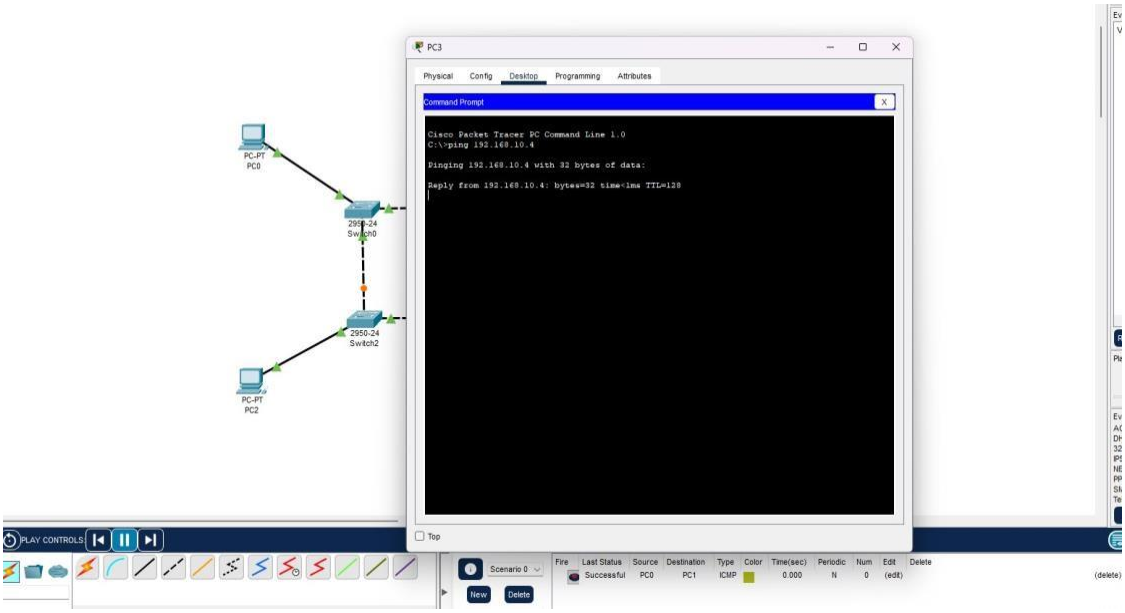


Figure 9. Command prompt of PC3 showing a ping reply from 192.168.10.4

Observation: The PDU from PC0 got to PC1 by way of Switch0 and Switch1. The link between Switch0 and Switch2 carries an orange dot, indicating that STP blocked it to avoid a loop while the remaining three links carry the traffic. The ping reply from 192.168.10.4 came back in less than 1 ms with TTL=128.

Result: The ring topology was built and connectivity between the PCs was confirmed, with STP preventing a switching loop.

4.5 Mesh Topology

Aim: To build a full mesh topology with four switches and confirm connectivity between the PCs.

Theory: In a full mesh topology each node has its own direct link to every other node. A network with n nodes requires n(n-1)/2 links, so four switches require six links. It offers the best fault tolerance because many alternative paths are available, but it is the costliest and most complicated topology to cable and maintain.

Devices Required

Device / Item	Type / Model	Quantity
PC-PT	End device	4
Cisco 2950-24 switch	Network device	4
Copper straight-through cable	Cable (PC to switch)	4
Copper cross-over cable	Cable (switch to switch)	6

Procedure

1. Put four 2950-24 switches (Switch0 to Switch3) and four PCs (PC0 to PC3) in the workspace.
2. Connect each switch to the other three switches using six copper cross-over cables.
3. Attach one PC to every switch with a copper straight-through cable.
4. Set the IP address and subnet mask on each PC.
5. Change to simulation mode, send a simple PDU from PC0 to PC3 and follow its path.
6. Open the Command Prompt on PC0 and ping 192.168.10.4.

IP Address Configuration

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.10.1	255.255.255.0
PC1	FastEthernet0	192.168.10.2	255.255.255.0
PC2	FastEthernet0	192.168.10.3	255.255.255.0
PC3	FastEthernet0	192.168.10.4	255.255.255.0

Screenshot

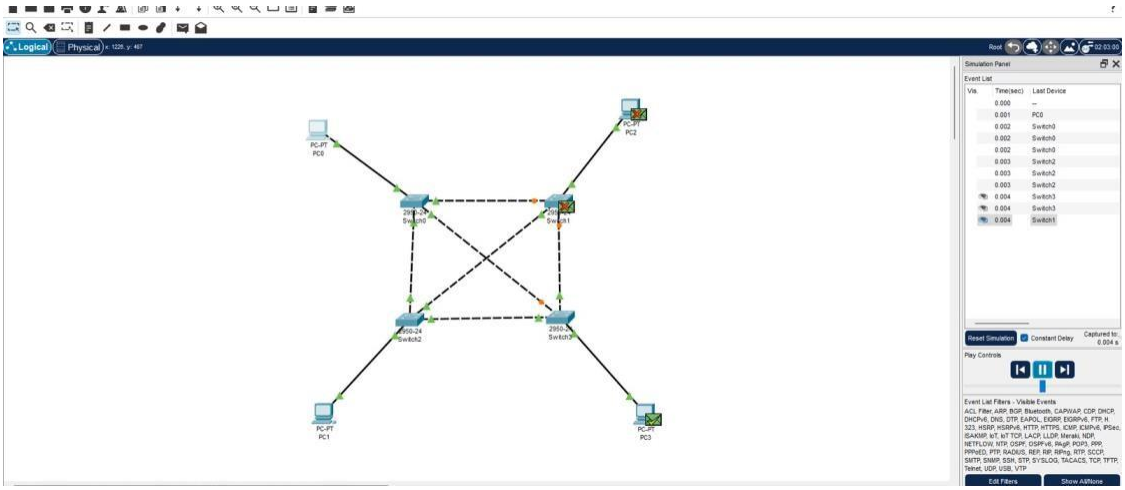


Figure 10. Mesh topology in simulation mode with a PDU from PC0 to PC3

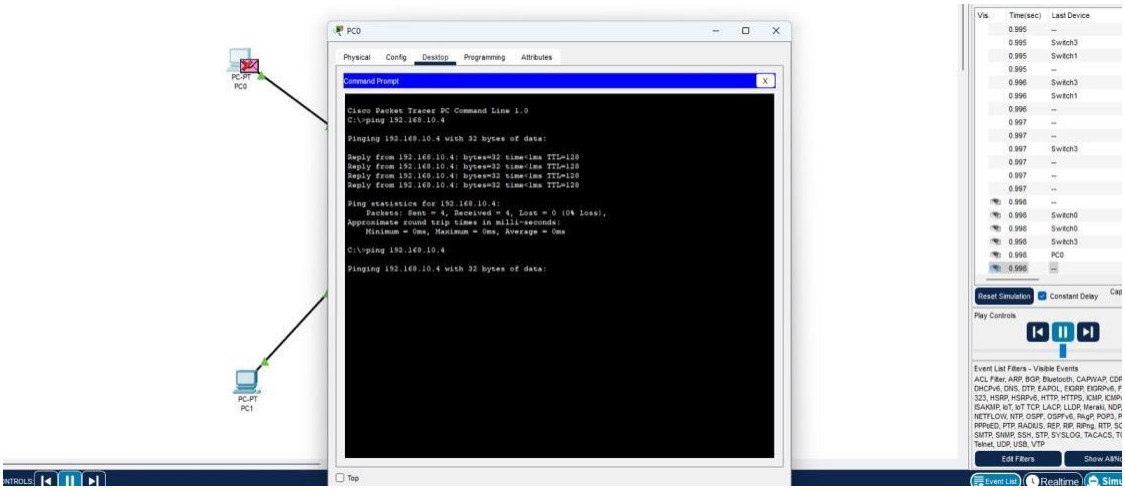


Figure 11. Command prompt of PC0 showing ping replies from 192.168.10.4

Observation: Every switch is directly linked to all the other switches. STP blocked some of the redundant links (orange dots) to prevent loops, and duplicate copies of the broadcast frame that reached other devices were discarded (red cross). The ping test transmitted 4 packets and received 4 replies with 0% loss; the round-trip time was under 1 ms with TTL=128.

Result: The mesh topology was built and connectivity was confirmed, with redundant links available between the switches.

4.6 Tree Topology

Aim: To build a tree topology with a root switch and two branch switches and confirm connectivity between the PCs.

Theory: A tree topology organises devices in a hierarchy: a root node is connected to branch nodes, which in turn serve the end devices. It blends the bus

and star topologies, can be expanded easily by adding more branches, and confines a fault to a single branch. If the root device fails, the branches lose contact with each other.

Devices Required

Device / Item	Type / Model	Quantity
PC-PT	End device	4
Cisco 2950-24 switch	Network device	3
Copper straight-through cable	Cable (PC to switch)	4
Copper cross-over cable	Cable (switch to switch)	2

Procedure

1. Put one root switch, two branch switches and four PCs in the workspace.
2. Join each branch switch to the root switch with copper cross-over cables.
3. Attach two PCs to each branch switch with copper straight-through cables.
4. Set the IP address and subnet mask on each PC.
5. Open the Command Prompt on a PC in one branch and ping a PC in the other branch.

IP Address Configuration

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.10.1	255.255.255.0
PC1	FastEthernet0	192.168.10.2	255.255.255.0
PC2	FastEthernet0	192.168.10.3	255.255.255.0
PC3	FastEthernet0	192.168.10.4	255.255.255.0

Screenshot

[Insert tree topology and command prompt screenshot here]

Observation: Traffic between PCs in the same branch is handled by that branch switch alone, whereas traffic between PCs in different branches has to pass through the root switch.

Result: The tree topology was built and connectivity between PCs in different branches was confirmed.

4.7 Hybrid Topology

Aim: To build a hybrid topology that merges ring, mesh, bus, star and hub-based segments and confirm connectivity between PCs in different segments.

Theory: A hybrid topology brings two or more basic topologies together in a single network. It is flexible and scalable, and each segment can use whichever topology suits it best, but it is harder to design, cable and manage. In this experiment a ring, a mesh, a bus, a star and a hub segment are connected to each other.

Devices Required

Device / Item	Type / Model	Quantity
PC-PT	End device	19
Cisco 2950-24 switch	Network device	13
Hub-PT	Network device	1
Copper straight-through cable	Cable (PC to switch or hub)	As required
Copper cross-over cable	Cable (switch to switch or hub)	As required

Procedure

1. Set up the ring segment using Switch0 to Switch3 and PC0 to PC3.
2. Set up the mesh segment using Switch4 to Switch7 and PC4 to PC7.
3. Set up the bus segment as a chain of Switch8 to Switch11 with PC15 to PC18.
4. Set up the star segment with Switch12 and PC11 to PC14, and the hub segment with Hub0 and PC8 to PC10.
5. Connect the segments using the switch-to-switch and switch-to-hub cross-over links and set the IP address and subnet mask on every PC.
6. Change to simulation mode, send a simple PDU from PC4 to PC15 and follow its path.
7. Open the Command Prompt on PC4 and ping 192.168.10.9.

IP Address Configuration

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.10.1	255.255.255.0

Device	Interface	IP Address	Subnet Mask
PC1	FastEthernet0	192.168.10.2	255.255.255.0
PC2	FastEthernet0	192.168.10.3	255.255.255.0
PC3	FastEthernet0	192.168.10.4	255.255.255.0
PC4	FastEthernet0	192.168.10.5	255.255.255.0
PC5	FastEthernet0	192.168.10.6	255.255.255.0
PC6	FastEthernet0	192.168.10.7	255.255.255.0
PC7	FastEthernet0	192.168.10.8	255.255.255.0
PC8	FastEthernet0	192.168.10.10	255.255.255.0
PC9	FastEthernet0	192.168.10.11	255.255.255.0
PC10	FastEthernet0	192.168.10.12	255.255.255.0
PC11	FastEthernet0	192.168.10.13	255.255.255.0
PC12	FastEthernet0	192.168.10.14	255.255.255.0
PC13	FastEthernet0	192.168.10.15	255.255.255.0
PC14	FastEthernet0	192.168.10.16	255.255.255.0
PC15	FastEthernet0	192.168.10.9	255.255.255.0
PC16	FastEthernet0	192.168.10.17	255.255.255.0
PC17	FastEthernet0	192.168.10.18	255.255.255.0
PC18	FastEthernet0	192.168.10.19	255.255.255.0

Screenshot

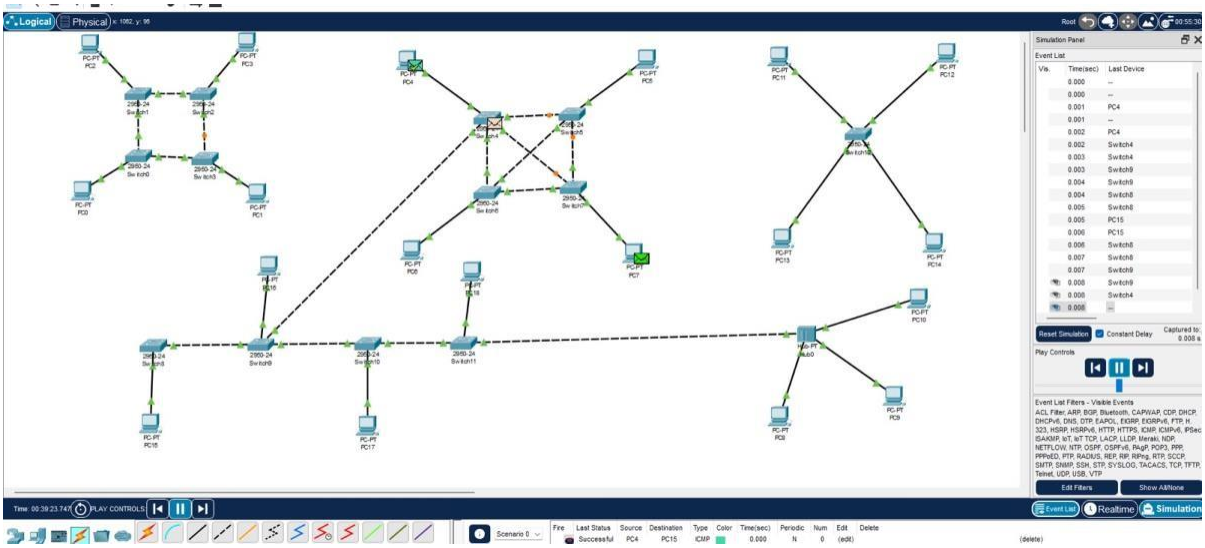


Figure 12. Hybrid topology in simulation mode with a PDU from PC4 to PC15

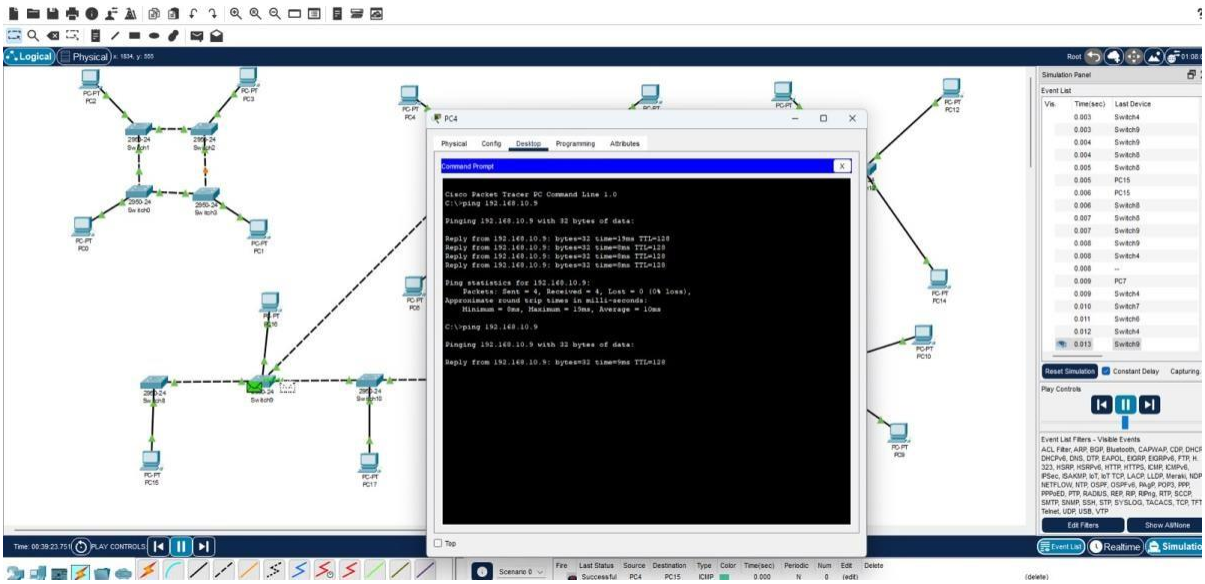


Figure 13. Command prompt of PC4 showing ping replies from 192.168.10.9

Observation: The PDU from PC4 in the mesh segment passed through Switch4, Switch9 and Switch8 to reach PC15 in the bus segment and came back along the same route. The ping test transmitted 4 packets and received 4 replies with 0% loss; the round-trip time ranged from 8 ms to 19 ms with an average of 10 ms and TTL=128. The delay is greater than in the single-topology networks because the packets cross several switches and segments.

Result: The hybrid topology was built and connectivity between PCs in different segments was confirmed.

5. COMPARISON OF NETWORK TOPOLOGIES

Topology	Connection Method	Cost	Fault Tolerance	Scalability	Main Limitation
Point-to-Point	Dedicated link between two devices	Very low	None; link failure ends communication	Not scalable	Only two devices
Bus	Single shared backbone	Low	Low; backbone break splits the network	Limited	Backbone failure and shared bandwidth
Star	Every node to a central switch	Medium	Node fault is isolated; switch fault stops the network	High	Central switch is a single point of failure
Ring	Each node to two neighbours in a closed loop	Medium	Medium; alternate path through the ring	Moderate	Needs STP to avoid loops
Mesh	Every node to every other node	Very high	Very high; many alternate paths	Difficult; n(n-1)/2 links	High cost and cabling complexity
Tree	Hierarchy of star networks under a root	Medium to high	Moderate; branch faults are isolated	High	Root failure cuts off the branches
Hybrid	Combination of two or more topologies	High	High	Very high	Complex design and management

6. OBSERVATION

- Point-to-point relies on one dedicated link and needs no switch, but it can serve only two devices.
- In the bus topology the PDU from PC0 to PC3 covered the entire backbone, and the ping average was 3 ms with 0% loss.
- In the star topology every packet went through the central switch, and the ping time was 3 ms to 4 ms with 0% loss.
- In the ring and mesh topologies STP blocked redundant links (orange dots) to avoid loops; the ping replies came back in under 1 ms.
- The tree topology sends traffic between branches through the root switch.
- In the hybrid topology the ping from PC4 to PC15 went across several segments, giving an average of 10 ms with 0% loss.
- Fault tolerance and cost increase together: the mesh is the most reliable and the costliest, while the point-to-point and bus topologies are the cheapest and the least reliable.

7. RESULT

The point-to-point, bus, star, ring, mesh, tree and hybrid topologies were built in Cisco Packet Tracer, connectivity in each was confirmed using ping and simulation mode, and the topologies were compared in terms of cost, fault tolerance and scalability.